

Theorem Statement

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation with associated matrix $A \in \mathbb{R}^{n \times n}$.

1. A is invertible
2. A has a pivot in every row
3. A has a pivot in every column
4. The system $A\mathbf{x} = \mathbf{b}$ has a unique solution for all $\mathbf{b} \in \mathbb{R}^n$
5. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
6. The columns of A are linearly independent
7. The columns of A span all of $\mathbb{R}^n$.
8. T is invertible
9. T is injective
10. T is surjective

We have the tools to prove all these, but really, it's just a summary of all the linear transformation and matrix algebra we've done so far!

One-sided inverse Implies the Other!

Theorem

Let $A \in \mathbb{R}^{n \times n}$. If there is some $B \in \mathbb{R}^{n \times n}$ such that

$$BA = I_n \text{ or } AB = I_n$$

Then, A is invertible and $A^{-1} = B$.

$AB = I_n$ case

$BA = I_n$ case